

EXPT NO:2	LOAD AND VISUALIZE A DATASET USING TABLEAU and PowerBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Why is Tableau a preferred tool for data visualization?

Tableau is preferred because it provides an easy drag-and-drop interface, interactive dashboards, and powerful data visualization capabilities without requiring programming skills.

2. What is the primary use of a line chart in data analysis?

A line chart is used to display trends and changes in data over time.

3. When would you use a tree map instead of a bar chart?

A tree map is used when comparing hierarchical data or showing the proportion of categories within a limited space.

4. What is the significance of filters in Tableau visualizations?

Filters allow users to focus on specific subsets of data, making analysis more interactive and meaningful.

5. How does Tableau handle geographic data for visualizations?

Tableau automatically recognizes geographic fields such as Country, State, and City and plots them on maps using built-in geographic coordinates.

IN-LAB EXERCISE:

OBJECTIVE:

To load a dataset into Tableau, create a variety of visualizations, and derive insights from the data using different chart types available in Tableau.

PROCEDURE:

Dataset Selection:

Use the Global Superstore Sales Dataset (available on Kaggle or Tableau's sample datasets).

Ensure the dataset includes fields like Order Date, Category, Region, Sales, Profit, and Customer Segment.

Data Loading:

Open Tableau and connect to the dataset.

Review the dataset structure, field data types, and missing values.

Chart Creations (Step-by-Step):

a) Bar Chart:

Objective: Compare total sales by region and category.

Plot Region on the x-axis and Sales on the y-axis.

Add Category as a color dimension.

b) Line Chart:

Objective: Visualize sales trends over time.

Place Order Date on the x-axis and Sales on the y-axis.

Set granularity to Month and color by Category.

c) Pie Chart:

Objective: Show sales distribution by product category. Assign

Category as the dimension and Sales as the measure.

d) Scatter Plot:

Objective: Examine the relationship between profit and sales.

Plot Profit on the x-axis and Sales on the y-axis.

Use Sub-Category as the color dimension.

e) Heatmap:

Objective: Compare sales across categories and regions.

Use Region and Category as dimensions. Use

Sales as the measure and apply color coding.

f) Geographical Map:

Objective: Visualize total sales by country.

Set Country as the geographical field. Use

Sales to define size and color gradients.

g) Tree Map:

Objective: Display sales distribution by category and sub-category.

Use Category and Sub-Category as dimensions, and Sales as the measure.

Customization and Interactivity:

Add filters to explore data subsets interactively (e.g., by year, region, or category).

Include annotations, labels, and formatting for clarity.

Dashboard Creation:

Combine all charts into a single dashboard to present a cohesive story.

Add interactive filters and tooltips for dynamic exploration.

Experiment Dataset:

Global Superstore Sales Dataset

Source: <https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

Fields Used:

Order Date

Region

Category

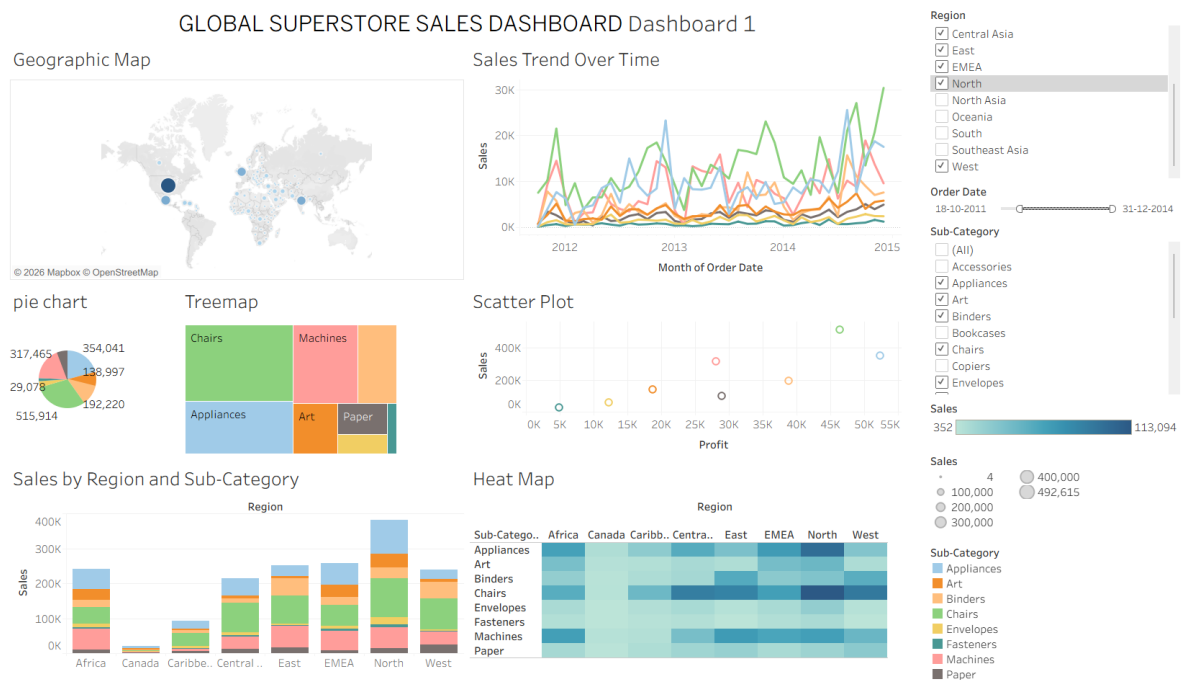
Sub-Category

Sales

Profit

Country

OUTPUT :



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

Bar Chart :

What regions had the highest and lowest total sales?

Highest Sales: Western Europe

Lowest Sales: Central Asia

How does the performance of categories vary across regions?

Technology generally has the highest sales across most regions, while Furniture and Office Supplies vary depending on the region.

Line Chart :

Identify the month with peak sales.

November recorded the highest sales.

Which category shows consistent growth in sales over time?

Technology shows the most consistent growth.

Pie Chart :

Which category contributes the most to overall sales?

Technology contributes the highest share of total sales.

Are there any categories with negligible sales contributions?

No. All three categories contribute significantly, although Furniture has the smallest share.

Scatter Plot :

Is there a noticeable correlation between profit and sales?

Yes. There is a positive correlation, but some high-sales products generate low or negative profit.

Which sub-category shows the highest sales but low profit?

Tables show high sales but relatively low or negative profit.

Heatmap :

Which region-category combination has the highest sales?

Western Europe – Technology has the highest sales.

Identify any regions with poor sales performance across all categories.

Central Asia shows relatively low sales across all categories.

Geographical Map :

Which country has the highest total sales?

United States has the highest total sales.

Are there regions with minimal or no sales?

Yes. Some smaller countries and regions have very low sales compared to major markets.

Tree Map :

Which sub-category dominates the sales in each category?

- Furniture: **Chairs**
- Office Supplies: **Storage**
- Technology: **Phones**

Are there any sub-categories with unexpectedly low sales?

Yes. Fasteners, Labels, and Envelopes have comparatively low sales.

Dashboard :**What insights can you derive when combining all visualizations?**

The dashboard provides a comprehensive view of regional sales, category performance, sales trends, profit versus sales relationships, geographical distribution, and top-performing sub-categories. It helps identify profitable markets, high-selling products, and areas that require improvement.

How do filters enhance the utility of the dashboard?

Filters allow users to analyze data by year, region, category, or country, making the dashboard interactive and enabling faster, more detailed decision-making.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		